

Course Project

Your course project score will be aggregated as follows:

- 20% proposal
- 25% video presentation
- 35% final report
- 20% peer evaluations given by your fellow group members

Proposal

Your project should be a **novel** proposal for a statistical application. You will be graded for originality.

For starters, your project should **not** be a direct adaptation of an existing project found online, e.g., a Kaggle project analysis that someone else has already done.

Your proposal needs to be submitted as a PDF file on gradescope, and should be no longer than 2 pages.

While formulating your project proposal, focus on the following:

1. Why is this problem important?
2. What are the data sources for this?
3. Where do you plan on acquiring the data from?
4. What type of analyses do you propose for the dataset?
5. What are your expected outcomes for the project?

Your project proposal grade will be based on the following rubric:

- **Clarity (40%)**

Is the proposal clear, well-organized, and easy to understand? Are the objectives and goals precisely defined?

- **Difficulty and Originality (40%)**

You will lose points if either: (i) the project is too simple and lacks originality, or (ii) does not provide a sufficient challenge for a ~3 person project.

- **Feasibility** (20%)

The proposal includes a realistic plan, and is feasible to do in the time frame of the course.

Final Report

Your final report should be a comprehensive document that details your project from start to finish. This means that you should describe all the steps you took in your project in a clear and concise manner. Include any tables, plots or other analyses/visualizations/results that you think are relevant. In a nutshell, your final report is supposed to be like a report you would submit in an English class but bolstered by scientific evidence and analyses.

The following is a general checklist for what to include in the final report.

- A statement of the problem you are investigating
- Why is this problem relevant? Or, what inspired you to investigate this problem?
- Where did you get the data from?
- Description of the data
- Exploratory data analysis
- What (if any) analyses have already been performed on this data (or another similar dataset)?
 - You should provide references to this
- What types of analyses did you perform?
- How do you interpret the results from these analyses?
- What are some potential limitations and shortcomings of your analyses?

⚠ Remember, not all projects need to end with a conclusive result. It is perfectly fine to have a project that ends with a "we need more data" or "we need to perform more analyses" conclusion.

However, you should be able to justify why you need more data or perform more analyses. This is where your statistical investigation skills come into play. ⚠️

Your final report will be graded based on the following rubric:

- **Clarity**

The final report needs to be clear, concise, and well-organized, making it easy for your TAs to understand the project's objectives, methods, and conclusions.

- **Exploratory Data Analysis**

Have you taken the time to understand your dataset? What are some interesting insights you have found? Are there missing values, outliers, or some nuisance variables? And if so, how did you handle them?

- **Statistical Analyses**

Are the statistical analyses you performed appropriate for the question you are investigating? Have the underlying assumptions been clearly stated and justified?

- **Interpretation of Results**

How do you interpret the results from your analyses? Are the results clearly presented and easy to understand? Are your interpretations consistent with the analyses you performed?

- **Relation to Project Proposal**

Does the final report relate to the initial proposal? Or, did you end up doing something completely different? If so, why? And how does this relate to the initial proposal?

- **Code execution**

You will be asked to submit your code along with your final report. Your code should be well-documented and easy to follow. Your TAs should be able to run your code without any issues and reproduce your results.

Video Presentation

In addition to your written report, your group is required to submit a recorded video presentation. This is your opportunity to "sell" your analysis and highlight the most interesting findings in an engaging format.

Requirements:

- **Format:** A video file (mp4, mov) uploaded directly to Canvas.
- **Length:** The video must be **less than 20 minutes** long.
- **Tools:** You may record your group's presentation using Zoom (i.e., start a new Zoom meeting with your group, share screen, and record to cloud/computer). Alternatively, you can also use OBS, Loom, or any other software of your choosing. You can use slides to structure your presentation. You can also convert your existing Jupyter notebook into RISE slides.

What your presentation should cover (suggested outline)

1. **Problem statement** and why it matters
2. **Data sources:** where you got the data, how you collected/cleaned it
3. **Exploratory Data Analysis (EDA):** key visuals + insights (missingness, outliers, key relationships)
4. **Statistical methods / inference:** what you did and why (assumptions stated clearly)
5. **Results:** main findings with interpretation (not just plots—explain what they mean)
6. **Limitations** and what you would do next with more time/data

Your presentation will be graded based on the following rubric:

- **Statistical communication (60%)**

Can you verbally explain your findings correctly? We are looking for precise interpretation of your EDA and results (e.g., correctly explaining coefficients, confidence intervals, or p-values). You should demonstrate that you understand *what* your results mean in the context of the real-world problem.

- **Presentation of results (40%)**

Does the presentation synthesize the project into a cohesive narrative, or is it just a reading of the written report? You should focus on the "highlights"—the most critical insights, the most interesting

outliers, and the key conclusions. Do not simply read text off your slides.

Peer Evaluations

You will be asked to evaluate your fellow group members at the end of the project. This evaluation will be used to adjust the final project grade.

They will grade you based on the following criteria:

- **Communication skills** (20%)

Assess how well the member communicated ideas, responded to questions, and engaged in discussions.

- **Meeting deadlines** (20%)

Evaluate the member's punctuality in finishing their assigned tasks by the agreed-upon deadlines.

- **Quality of work** (20%)

Assess the accuracy, thoroughness, and overall quality of work produced by the group member

- **Overall Contribution to the project** (20%)

Consider the group member's overall involvement, participation and efforts towards completing the project objectives

- **Honesty in evaluation** (20%)

How well does your own evaluation of your contribution to the project (based on the above 4 criteria) agree with your group members' evaluations

Topics from past projects

To get a better idea of what you can work on, here are some topics from past projects.

1. Analyzing the distribution of staff wages and potential disparities between departments at the University of California.
2. Investigating the factors that influence sleep quality and duration, and how they differ between occupations.
3. Predicting the presence of sleep disorders based on various health markers.
4. Analyzing the relationship between housing affordability and different factors in San Diego.
5. Examining the impact of the COVID-19 pandemic on elementary school literacy rates in California.
6. Investigating the geographical and temporal distribution of Nobel Prize recipients and factors that contribute to winning a Nobel Prize.
7. Analyzing the relationship between federal government spending on pharmaceutical industries and drug usage among the American population.
8. Quantifying urban land expansion using satellite data and identifying factors that influence it.
9. Performing statistical inference on U.S. presidential election data to understand political preferences across states and over time.
10. Examining the factors that contribute to the adoption of clean energy sources in different countries.
11. Investigating the relationship between household housing cost burdens and educational attainment in California.
12. Analyzing the impact of various game characteristics on success in the Steam marketplace.
13. Examining the differences in physical fitness among students of different ethnic groups and socioeconomic regions in San Diego County.
14. Assessing the transferability of a college basketball player's skills to the low-Division 1 level, such as UC San Diego.